

KIARASH ALIREZAEI

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EDUCATION

University of Toronto, St. George

Toronto, Ontario

Bachelor of Applied Science in Computer Engineering (GPA: 3.74/4.0)

Sep 2022 – May 2027

Minors and Certificates: Artificial Intelligence Engineering Minor, Engineering Business Minor, Cybersecurity Certificate

Relevant Coursework: Algorithms and Data Structures, Operating Systems (C), Distributed Systems (Go), Computer Networks, Introduction to Machine Learning, Introduction to Artificial Intelligence, Applied Fundamentals of Deep Learning (Python), Probability & Applications, Computer Security, Digital Systems (Verilog), Digital Electronics, Computer Organization (Assembly)

EXPERIENCE

University of Toronto

Sep 2026 – Present

Applied AI Research Engineering Intern

Toronto, Ontario

- Co-leading development of audio and visual machine learning models for real-time commercial detection and automated advertisement replacement.
- Developing data processing and inference pipelines connecting Raspberry Pi devices, server-side ML models, and backend services.
- Contributing to ML systems that translate visual interface content into interactive tactile representations for blind and low vision users.

IBM Canada

May 2025 – Aug 2026

Compiler Developer Intern – IBM Python for z/OS

Markham, Ontario

- Enhanced IBM Open Enterprise SDK for Python using C and Python, integrating z/OS system functions into CPython and debugging compiler and runtime compatibility issues.
- Investigated customer-reported import, package installation, and application issues; contributed fixes to upstream Python projects including distutils.
- Automated download, installation, and validation of IBM Python AI Toolkit pip packages on z/OS; authored technical blogs on Python libraries, tools, installation, and debugging.
- Integrated upstream libraries such as zstd and maintained automated test and CI pipelines using Git and Buildbot to support reliable Python builds and regression testing on z/OS.

Reviewerly

May 2024 – Aug 2024

Software Engineer Intern

Toronto, Ontario

- Developed RESTful APIs and backend functionality using Java and MongoDB, supporting data creation, retrieval, and application workflows.
- Migrated search functionality from MongoDB queries to Apache Solr, reducing search response times by over 50%.
- Built responsive React interfaces and integrated backend APIs using React Query, improving state management, data visualization, and user interaction.

PROJECTS

Fault-Tolerant Distributed Key-Value Store | Go, Raft, RPC, Distributed Systems

Jan 2025 - Apr 2025

- Implemented the Raft consensus protocol in Go, including randomized leader election, heartbeat-based failure detection, replicated logs, majority-based commits, and automatic recovery from node failures.
- Developed persistent state and optimized log-reconciliation mechanisms to recover crashed replicas, resolve conflicting logs, and efficiently resynchronize nodes after network partitions.
- Built a linearizable distributed key-value store on top of Raft, supporting concurrent Get/Put operations, client retries, leader changes, server restarts, and unreliable network conditions.

Skin Lesion Classifier (NLC7) | Python, PyTorch, Matplotlib, CNNs, Transfer Learning

May 2024 – Aug 2024

- Processed the large-scale HAM10000 dataset, applying advanced data augmentation techniques using Pandas and Torch libraries to balance 6000 images across cancer types and ensure robust model training.
- Designed a deep learning pipeline incorporating U-Net for precise image segmentation, followed by ResNet-50 for feature extraction and MLP for classification, enabling accurate identification of skin cancer types.
- Researched state-of-the-art AI image processing techniques and applied hyperparameter tuning across U-Net, CNNs, and MLP models, achieving 92% accuracy while reducing RAM usage by 75% and optimizing model performance.

Interactive Maps Application Development | C++ (STL), EZGL, GTK

Feb 2024 – Mar 2024

- Developed an interactive C++ mapping application with real-time data APIs, a responsive interface, and route optimization using A*, Multi-Dijkstra's, and a Traveling Salesman Problem solution.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Go, JavaScript, SQL, Bash/Shell, Verilog, Assembly, MATLAB

Systems & Platforms: CPython, z/OS USS, Linux, POSIX APIs, Systems Programming, Compiler & Runtime Development, IBM Clang, Distributed Systems, Concurrency, Fault Tolerance, RPC, REST APIs

AI & Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, Deep Learning, Neural Networks, Transfer Learning, Computer Vision

Frameworks & Software: React, React Native, Spring Boot, Django, GTK, EZGL

Hardware & Digital Systems: FPGA (DE1-SoC), Quartus, ModelSim, VGA, PS/2, Finite State Machines, Datapaths, Digital Logic

Cloud & DevOps: AWS, Git, Buildbot, Ansible, Docker, CI/CD, pytest, Test Automation, Python Packaging, pip, Bandit, Detect-Secrets

Databases & Search: MongoDB, MySQL, Apache Solr